

KUANG-CHENG, CHENG

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EDUCATION

- Sep. 2021-
May. 2023 **University of Wisconsin-Madison, Madison**
M.S. in Geographic Information Systems and Cartography
- GPA: 3.83
- Sep. 2011-
Jun. 2015 **National Central University, Taoyuan, Taiwan**
B.S. in Civil Engineering
- EAC accredited program.
 - GPA: 3.34.
 - Relevant coursework:
 - (1) Programming for Engineering (C++) using Visual Studio.
 - (2) Numerical Analysis: MATLAB Program design.
 - (3) Engineering Statistics.

WORK EXPERIENCE

- Apr. 2024-
Now **GIS Analyst**
Houston Public Works *Houston, TX*
- Leverage ModelBuilder to optimize geoprocessing workflows, speed up the processing time, and reduce human error.
 - Create, maintain, update geospatial data on ArcGIS enterprise database.
 - Perform data QA/QC to ensure the accuracy and quality of geospatial data
 - Design and create maps to support engineering projects.
 - Coordinate and manage the GIS projects for engineering and IT groups to ensure data integrity and determine project priorities.
- Sep. 2023-
Apr. 2024 **GIS Volunteer**
URISA's GISCorps *Des Plaines, IL*
- Test and help with the development of the user interface (UI) of Underpass, a ReactJS library, for the Humanitarian OpenStreetMap Team.
 - Collect street-level images in support of the creation of smart map data for Mapilio, a map platform.
 - Edit buildings on satellite photos for Humanitarian OpenStreetMap Team projects.
- Jan. 2022-
Aug. 2022 **GIS Metadata Technician**
University of Wisconsin-Madison *Madison, WI*
- Assessed LiDAR data inventory statuses.
 - Created original ISO metadata for nearly 400 new Wisconsin county datasets.
 - Added interactive tile indexes for Wisconsin LiDAR-derivative data resources.
 - Edited Wisconsin V8 county parcel data.
- Aug. 2019-
Jul. 2021 **Assistant Engineer**
Ministry of Transportation and Communication *Taipei, Taiwan*
- Coordinated the COVID-19 mask distribution project to ensure companies and agencies had sufficient supplies to maintain workplace safety.
 - Processed COVID-19 mask demand data and monitored logistics to ensure timely and accurate distribution.
 - Emergency Response Duty: Participated in on-call shifts for natural disasters (e.g., earthquakes, typhoons) and mobility-related incidents, ensuring communication and support during emergency response efforts.
 - Updated, maintained, and delivered routine mobilization data.
 - Evaluated mobilization task guidelines for air, shipping, and road transportation.
 - Arranged the annual training program for senior officers and project leaders.
- Nov. 2016-
Jul. 2021 **Junior Engineer**
Freeway Bureau *New Taipei, Taiwan*

- Review and revise environmental impact assessment reports, ensuring they are aligned with policies and regulations.
- Digitized Taiwan's freeway historical documents for historical data preservation.
- Held environmental education events, evoking awareness of environmental issues.
- Examined design drafts and contracts of national bridges.

**Jul. 2014-
Aug. 2014** **Engineer Intern**
CECI Engineering Consultants, Inc. *Taipei, Taiwan*
 Gained experience at the most prominent civil engineering design company in Taiwan.

- Visualized civil engineering project drafts and data on Google Earth.
- Completed sectional drawings of roads on the mountain area of Tai-20 (one of Taiwan's provincial highways) using AutoCAD.

SKILLS

Languages: Mandarin Chinese (native), English (fluent).

Computer Software: AutoCAD, SketchUp, ArcGIS Pro, QGIS, MS Power BI, Adobe Illustrator, MS Office (including Word, Excel, PowerPoint)

Programming: Python, SQL, HTML/CSS/ JavaScript, GitHub

Cloud Computing Environment: Google Cloud Platforms, ArcGIS Online (AGOL)

Database System: PostgreSQL (with PostGIS), MongoDB, Snowflake

Other: Multi-tasking, Time Management, Problem-solving, Project Coordination, Supervisory Skills, Data Management, Communication

GIS PROJECTS

- **Spatial Big Data Analysis & Visualization:**
 "Madison Metro Transit Redesign Evaluation" - Perform data ETL, statistical data analysis and visualize spatial data on an interactive map using Python in Jupyter notebook.
- **Spatial Database Design and Management:**
 "US National Parks Recreational Database" - Create database to integrate visitation records, facilities, and spatial data inside or nearby national parks using PostgreSQL (with PostGIS extension).
- **GIS Analysis:**
 "Road salt use impact on major sensitive areas in the metropolitan area of Marathon County"-Perform spatial analysis and map the potential contamination strength on each road for marathon county transportation planner.
- **Geovisualization and UI/UX Design:**
 "United States Exports By Industry" - Create a web-based user interface to visualize the data collected from the Observatory of Economic Complexity (OEC) using JavaScript/HTML/CSS with Leaflet library.
- **GIS Development Practicum:**
 "Yearly ridership data analysis and visualization" - create a workflow for data ETL, open data publishing, and spatial data management and visualizations.
- **Cartography (Mapping) Design:**
 "Taiwan Public Transportation Infrastructure – bus ridership vs. population"-Aggregate data from multiple open data resources and visualize data on a map using ArcGIS Pro and Adobe Illustrator.

CERTIFICATION

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| 2016 | Professional and Technical Personnel Examinations of Civil Engineers, Taiwan <ul style="list-style-type: none"> • Certificate issued by Ministry of Examination, R.O.C. (Taiwan). • Officially qualified to be a Professional Civil Engineer (PE) in Taiwan. • Qualification Rate (2016): 11.53%. |
| 2016 | Civil Service Level Three Senior Examination in Civil Engineering, Taiwan <ul style="list-style-type: none"> • Certificate issued by Ministry of Examination, R.O.C. (Taiwan). • Passed the public servant exam as a Junior Engineer. • Qualification Rate (2016): 25.78%. |